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Alternating Direction Method of Multipliers Based Coordination Control of Multi-Vehicles and Traffic Signal*

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***Abstract*—This research proposes a coordination method for multi-connected and automated vehicles (CAVs) and traffic signal. It aims at reducing stop-and-go maneuvers of CAVs and enhancing traffic efficiency. The proposed method has the following highlights: i) Adaptive to actual CAV and human-driven vehicle (HV) mixed traffic; ii) Jointly optimization of both vehicle trajectory and signal timing via formulating in the spatial domain; iii) Parallel distributed computing. Simulation test results demonstrate that the proposed coordinated control significantly outperforms the benchmark method. The proposed method reduces the average travel delay by 29.56%, enhances fuel efficiency by 18.87%, and reduces stop count by 87.10%. The proposed parallel distributed computing algorithm ensures a computation time basically within 10 milliseconds. It indicates that the proposed method is ready for real-time large-scale implementation.**

I. Introduction

The signalized intersection represents a key bottleneck in urban transportation systems, resulting in a 5% to 10% increase in driving delay [1] and a 14% rise in emissions and fuel consumption [2]. In response to these challenges, the concept of vehicles-signal coordination has been proposed. It coordinates the trajectories of connected and automated vehicles (CAVs) and the timing plan of traffic signal, with an objective of mitigating stop-and-go maneuvers and enhancing traffic capacity [3-5].

However, there are still great limitations of current vehicle-signal coordination methods. Firstly, most studies are not adaptive to the CAV and human vehicle (HV) mixed traffic [6]. Secondly, conventional studies are lack of global optimality due to the decoupled modeling of traffic signal control and vehicle trajectory planning [7]. Thirdly, computation efficiency is a concern for large-scale implementation [8-10].

To address these problems, this study presents a coordination method of CAVs and signal. It bears the following features:

- Coordinating CAVs with traffic signal in a CAV and HV mixed environment;
- Optimizing CAVs' trajectories while preserving global traffic mobility;
- With large-scale implementation capability using alternating direction method of multipliers (ADMM) algorithm.

II. Methodology

A. Highlights

- Adaptive to actual CAV and HV mixed traffic.
- Jointly optimization of both vehicle trajectory and signal timing via formulating in the spatial domain [11-13].
- Parallel distributed computing with ADMM algorithm.

B. System Structure

The computation network is composed of three types of controllers in this study:

- **CAV controller**: The computing nodes associated with CAVs are referred to as CAV controllers.
- **Link controller**: The link controller is to avoid collisions between two adjacent vehicles.
- **Signal controller**: The signal controller is for optimizing signal timing plan.

C. Problem Formulation

The coordinated control problem is firstly formulated as a centralized formulation (1):

$$\min J = \sum_{i \in V_{\mathrm{A}}} J_i(u_i, G) + J_{\mathrm{s}}(\{u_i\}_{i \in V}, G)$$

subject to: (1)

$$(u_i, G) \in \mathbb{C}_{\mathrm{A}}, i \in V_{\mathrm{A}}$$

$$(u_i, G) \in \mathbb{C}_{\mathrm{C}}, i \in V_{\mathrm{A}}$$

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$$G \in \mathbb{C}_{\mathrm{S}}$$

where $V_{\mathrm{A}}, V_{\mathrm{H}}$ and V denote the set of CAVs, HVs, and all vehicles, respectively.

1) Decision Variables

The decision variable of CAV i is $u_i = [t_i, v_i]$, where $t_i = \left[t_{i,x_{i,0}}, t_{i,x_{i,0}+1}, \dots, t_{i,x_L}\right]$ and $v_i = \left[v_{i,x_{i,0}}, v_{i,x_{i,0}+1}, \dots, v_{i,x_L}\right]$ are the arrival time and speed of vehicle i in the spatial domain. $x_{i,0}$ is the initial position of CAV i and x_L is the position of stop line. $G = \left\{\left[t_p^{\mathrm{s}}, t_p^{\mathrm{e}}\right]\right\}_{p \in P}$ is the control variable of signal timing, where t_p^{s} and t_p^{e} denote the start and end time of green duration of phase p. P is the set of all phases of the signal plan. $\{u_i\}_{i \in V}$ denotes the arrival time and speed (in the spatial domain) of all vehicles in set V.

2) Objective Function

The objective for CAV is shown in Eq. (2). The reference path u_i^{ref} is the trajectory with free flow speed.

$$J_i(u_i, G) = \beta_1 \sum_{x=x_{i,0}}^{x_L+1} \left\|u_{i,x} - u_{i,x}^{\mathrm{ref}}\right\|_2^2 + \beta_2 \sum_{x=x_{i,0}+1}^{x_L+1} \left|a_{i,x}\right| \tag{2}$$

The objective of signal timing plan is to minimize the total delay of all HVs and CAVs at the intersection:

$$J_{\mathrm{s}}(\{u_i\}_{i \in V}, G) = \sum_{i \in V} \left(t_{i,x_L+1} - t_{i,x_L+1}^{\mathrm{ref}}\right) \tag{3}$$

where t_{i,x_L+1} is the actual arrival time of vehicle i pass the stop line (at $x_L + 1$).

3) CAV Dynamic Constraints in the Spatial Domain

We model the CAV dynamic constraints $\mathbb{C}_{\mathrm{A}}$ in the spatial domain as follows:

$$t_{i,x+1} = t_{i,x} + \frac{\Delta x}{v_{i,x}}, \quad x = x_{i,0}, \dots, x_L \tag{4}$$

$$v_{i,x+1}^2 = v_{i,x}^2 + 2\Delta x \cdot a_{i,x}, \quad x = x_{i,0}, \dots, x_L \tag{5}$$

$$a_{\min} \le a_{i,x} \le a_{\max}, \quad x = x_{i,0}, \dots, x_L, x_L + 1 \tag{6}$$

$$0 < v_{i,x} \le v_{\max}, \quad x = x_{i,0}, \dots, x_L, x_L + 1 \tag{7}$$

$$f\left(t_{i,x_L}, G\right) = t_{i,x_L} \tag{8}$$

Constraints (4) and (5) represents vehicle dynamics in the spatial domain. Constraints (6)-(7) limit the reasonable range of $a_{i,x}$ and $v_{i,x}$. Constraint (8) ensure CAV pass the intersection without stop. We denote $f\left(t_{i,x_L}, G\right)$ as the function that finds the start of next green time [13]. $f\left(t_{i,x_L}, G\right) = t_{i,x_L}$ indicates that time t_{i,x_L} is in a green phase.

4) Collision Avoidance Constraint

$\mathbb{C}_{\mathrm{C}}$ is the constraint set to avoid collision:

$$t_{i,x} - t_{j,x} \ge t_{\mathrm{safe}}, \forall e \in \mathcal{E}, (i,j) \in e, i \in V_{\mathrm{A}} \tag{9}$$

where $\mathcal{E}$ is the total set of all links of adjacent pairwise vehicles. i is the following vehicle and j is the leading vehicle.

5) Signal Timing Constraints

$\mathbb{C}_{\mathrm{S}}$ denotes the set of signal timing constraints:

$$g_{\min} \le t_p^{\mathrm{e}} - t_p^{\mathrm{s}} \le g_{\max}, \forall p \in P \tag{10}$$

$$t_{p+1}^{\mathrm{s}} = t_p^{\mathrm{e}} + l, \forall p \in P \tag{11}$$

$$\sum_{p \in P} \left(t_p^{\mathrm{e}} - t_p^{\mathrm{s}}\right) + L = C \tag{12}$$

where $g_{\min}$ and $g_{\max}$ denote the minimum and maximum green duration. l is the loss time and inter-green time. L is the total loss time. C is the cycle length.

6) HV Motion Prediction

The estimated location and speed of HVs are needed for optimizing the signal timing parameters and trajectory of CAVs [14]. Hence, the future trajectories of HVs are predicted based the intelligent driver model (IDM) [15].

D. Parallel Solution Algorithm

We use indicator functions $\mathbb{I}_{\mathrm{A}}(\cdot)$, $\mathbb{I}_{\mathrm{C}}(\cdot)$ and $\mathbb{I}_{\mathrm{s}}(\cdot)$ to describe constraints of CAV, pairwise interactive vehicles, and signal timing:

$$\mathbb{I}_{\mathrm{A}}(u_i, G_i) = \begin{cases} 0, if\ (u_i, G_i) \in \mathbb{C}_{\mathrm{A}}, i \in V_{\mathrm{A}} \\ +\infty, \quad otherwise \end{cases} \tag{13}$$

$$\mathbb{I}_{\mathrm{C}}\left(\{u_i^e\}_{i \in V(e)}\right) = \begin{cases} 0, if\ \left(\{u_i^e\}_{i \in V(e)}\right) \in \mathbb{C}_{\mathrm{C}} \\ +\infty, \quad otherwise \end{cases} \tag{14}$$

$$\mathbb{I}_{\mathrm{s}}(G_{\mathrm{s}}) = \begin{cases} 0, \quad if\ G_{\mathrm{s}} \in \mathbb{C}_{\mathrm{S}} \\ +\infty, \quad otherwise \end{cases} \tag{15}$$

By introducing consensus constraints, problem (1) could be rewritten into the consensus optimization problem [16]:

$$\begin{aligned} \min \hat{J} = & \sum_{i \in V_{\mathrm{A}}} \underbrace{\left[J_i(u_i, G_i) + \mathbb{I}_{\mathrm{A}}(u_i, G_i)\right]}_{\text{CAV objective}} \\ & + \sum_{e \in \mathcal{E}} \underbrace{\mathbb{I}_{\mathrm{C}}\left(\{u_i^e\}_{i \in V(e)}\right)}_{\text{Link objective}} + \underbrace{J_{\mathrm{s}}(\{u_i^{\mathrm{s}}\}_{i \in V}, G_{\mathrm{s}}) + \mathbb{I}_{\mathrm{s}}(G_{\mathrm{s}})}_{\text{Signal objective}} \end{aligned} \tag{16}$$

subject to:

$$u_i = z_i^u, u_i^e = z_i^u, u_i^s = z_i^u, i \in V_A, e \in \mathcal{E}(i)$$

$$G_i = z^G, G_s = z^G, i \in V_{\mathrm{A}}, e \in \mathcal{E}(i)$$

where z_i^u and z^G are consensus variables of vehicle i's control variable and signal timing plan, respectively.

The scaled augmented Lagrangian function is:

$$\begin{aligned} L_\rho = & \sum_{i \in V_{\mathrm{A}}} \left[J_i(u_i, G_i) + \mathbb{I}_{\mathrm{A}}(u_i, G_i)\right] + \sum_{e \in \mathcal{E}} \mathbb{I}_{\mathrm{C}}\left(\{u_i^e\}_{i \in V(e)}\right) \\ & + J_{\mathrm{s}}(\{u_i^{\mathrm{s}}\}_{i \in V}, G_{\mathrm{s}}) + \mathbb{I}_{\mathrm{s}}(G_{\mathrm{s}}) \\ & + \underbrace{\sum_{i \in V_{\mathrm{A}}} \frac{\rho}{2} \|u_i - z_i^u + \lambda_i^u\|_2^2 + \sum_{i \in V_{\mathrm{A}}} \frac{\rho}{2} \|G_i - z^G + \lambda_i^G\|_2^2}_{\text{Lagrangian function of CAV}} \\ & + \underbrace{\sum_{e \in \mathcal{E}} \sum_{i \in V(e)} \frac{\rho}{2} \left\|u_i^e - z_i^u + \lambda_i^{u,e}\right\|_2^2}_{\text{Lagrangian function of link}} \\ & + \underbrace{\sum_{i \in V} \frac{\rho}{2} \left\|u_i^{\mathrm{s}} - z_i^u + \lambda_i^{u,s}\right\|_2^2 + \frac{\rho}{2} \|G_{\mathrm{s}} - z^G + \lambda_{\mathrm{s}}^G\|_2^2}_{\text{Lagrangian function of signal}} \end{aligned} \tag{17}$$

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where λ_i^u, $\lambda_i^{u,e}$, $\lambda_i^{u,s}$, λ_i^G, and λ_s^G are scaled dual variables, and ρ is the penalty parameter. The ADMM algorithm minimizes the augmented Lagrangian from with three steps iteratively:

Step 1: Each CAV controller $i \in V_A$ solves the subproblem:

$$u_i^{k+1}, G_i^{k+1} = \operatorname{argmin}_{u_i, G_i} \left\{ [J_i(u_i, G_i) + \mathbb{I}_A(u_i, G_i)] + \frac{\rho}{2} \left\| u_i - z_i^{u,k} + \lambda_i^{u,k} \right\|_2^2 + \frac{\rho}{2} \left\| G_i - z^{G,k} + \lambda_i^{G,k} \right\|_2^2 \right\} \tag{18}$$

Each link controller $e \in \mathcal{E}$ solves the problem:

$$\{u_i^{e,k+1}\}_{i \in V(e)} = \operatorname{argmin}_{u_i^e} \left\{ \mathbb{I}_C(\{u_i^e\}_{i \in V(e)}) + \sum_{i \in V(e)} \frac{\rho}{2} \left\| u_i^e - z_i^{u,k} + \lambda_i^{u,e,k} \right\|_2^2 \right\} \tag{19}$$

The signal controller solves the subproblem:

$$\{u_i^{s,k+1}\}_{i \in V}, G_s^{k+1} = \operatorname{argmin}_{u_i^s, G_s} \left\{ J_s(\{u_i^s\}_{i \in V}, G_s) + \mathbb{I}_s(G_s) + \sum_{i \in V} \frac{\rho}{2} \left\| u_i^s - z_i^{u,k} + \lambda_i^{u,s,k} \right\|_2^2 + \frac{\rho}{2} \left\| G_s - z^{G,k} + \lambda_s^{G,k} \right\|_2^2 \right\} \tag{20}$$

Step 2: Update the consensus variables z_i^u and z^G:

$$z_i^{u,k+1} = \frac{(u_i^{k+1} + \lambda_i^{u,k}) + \sum_{e \in \mathcal{E}(i)} (u_i^{e,k+1} + \lambda_i^{u,e,k}) + (u_i^{s,k+1} + \lambda_i^{u,s,k})}{1 + \sum_{e \in \mathcal{E}(i)} 1 + 1} \tag{21}$$

$$z^{G,k+1} = \frac{\sum_{i \in V_A} (G_i^{k+1} + \lambda_i^{G,k}) + (G_s^{k+1} + \lambda_s^{G,k})}{\sum_{i \in V_A} 1 + 1} \tag{22}$$

Step 3: The scaled dual variable λ is updated:

$$\lambda_i^{u,k+1} = \lambda_i^{u,k} + (u_i^{k+1} - z_i^{u,k+1}) \tag{23}$$

$$\lambda_i^{u,e,k+1} = \lambda_i^{u,e,k} + (u_i^{e,k+1} - z_i^{u,k+1}) \tag{24}$$

$$\lambda_i^{u,s,k+1} = \lambda_i^{u,s,k} + (u_i^{s,k+1} - z_i^{u,k+1}) \tag{25}$$

$$\lambda_i^{G,k+1} = \lambda_i^{G,k} + (G_i^{k+1} - z_i^{G,k+1}) \tag{26}$$

$$\lambda_s^{G,k+1} = \lambda_s^{G,k} + (G_s^{k+1} - z_i^{G,k+1}) \tag{27}$$

III. Evaluations

A. Experiment Design

The proposed method is implemented in a typical four-arm intersection by simulation. Simulations were implemented in SUMO, an open-source traffic simulation software [17]. The simulation is controlled through the interface *TraCI* in Python 3.11 with *Gurobi* as a solver in an Intel i9-13900H processor at 3.4 GHz and 32 GB of RAM.

B. Experiment Results

1) Function Validation

Fig. 1 shows that the signal controller can adjust timing plan according to traffic conditions. The CAV can plan a smooth trajectory to pass the intersection without stop. CAVs can also lead following HVs to pass the intersection smoothly, improving overall operational efficiency at the intersection.

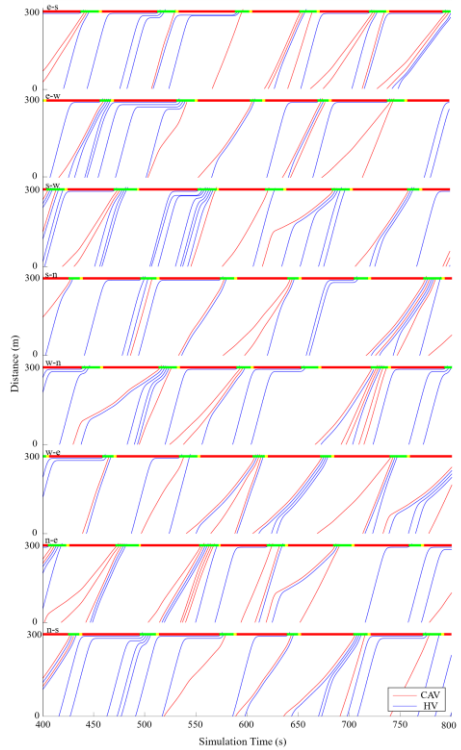


Figure 1. Trajectories of CAVs and HVs

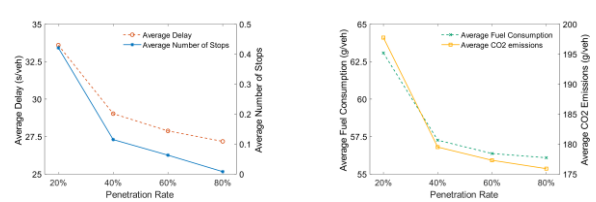
2) Mobility and Sustainability

As shown in Table 1, the coordinated control can significantly reduce travel delay by 29.56%. The number of stops is reduced remarkably (by 87.10%). The fuel consumption and CO2 emission for all vehicles are significantly decreased (by 18.87%).

TABLE I. Control Performance Comparison

	Average delay (s/veh)	Average number of stops	Average Fuel Consumption (g/veh)	Average CO2 emissions (g/veh)
Actuated	41.21	0.93	70.58	221.28
Proposed	29.03	0.12	57.26	179.51

Fig. 2(a) shows that the global mobility is enhanced with the increase in CAV penetration rate. Fuel consumption and CO2 emissions are also reduced as shown in Fig. 2(b). The marginal reductions in both mobility and sustainability are the most significant when the penetration rate increases from 20% to 40%.



(a) Mobility performance (b) Sustainability performance

Figure 2. Sensitive analysis under different CAV penetration rates

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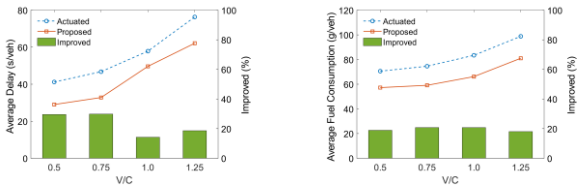
Fig. 3(a) and Fig. 3(b) demonstrate the superior performance of our proposed method across all demand conditions (from under-saturated (V/C=0.5) to over-saturated (V/C=1.25)) compared to the actuated control.

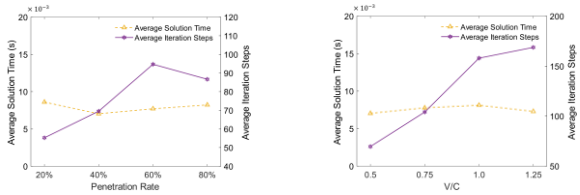


(a) Mobility performance (b) Sustainability performance

Figure 3. Sensitive analysis under different demand levels

3) Computational Efficiency

As shown in Fig. 4(a) and Fig. 4(b), the average solution time remains almost constant across all CAV penetration rates and V/C ratios. The average iteration number slightly increases with the increase of CAV number. Considering the short solution time of each subproblem, the algorithm still holds great significance in large-scale applications.



(a) CAV Penetration rate (b) Demand level

Figure 4. Sensitive analysis of computational efficiency

IV. Conclusion

This research proposes a coordination method for multi-connected and automated vehicles (CAVs) and traffic signal. The proposed method has the following highlights: i) Adaptive to actual CAV and HV mixed traffic; ii) Jointly optimization of both vehicle trajectory and signal timing via formulating in the spatial domain; iii) Parallel distributed computing. Evaluations show that:

- The proposed coordinated optimization model can significantly reduce delay by 29.56%, improve fuel efficiency by 18.87%, and reduce stop count 87.10% compared to the vehicle-actuated control method.
- In the parallel ADMM algorithm, the solution time of each sub-problem is basically within 10ms.
- The proposed method demonstrates good performance across various CAV penetration rates and demand levels.

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